emphasis: Evolutionary Modeling for Phylogenetic Inference

Technical Reference Manual

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Contents

1 Notation **3**

2 The Diversification Model **3**
2.1 General rate specification . . . 3
2.2 Special cases (model shortcuts) . . . 4

3 The Phylogenetic Likelihood **5**
3.1 Observed versus augmented data . . . 5
3.2 Complete-data log-likelihood . . . 5
3.3 The proposal distribution . . . 6

4 Importance-Sampling Likelihood Estimation **6**
4.1 The IS estimator . . . 6
4.2 Bias-corrected estimator . . . 7
4.3 Effective Sample Size . . . 8
4.4 Cross-importance sampling (shared-tree mode) . . . 8

5 The Cross-Entropy Method for Parameter Estimation **8**
5.1 Overview . . . 8
5.2 Algorithm . . . 8
5.3 Stopping rules . . . 9
5.4 Particle rescue under limit exceedance . . . 10
5.5 Annealing schedule . . . 10

6 Monte Carlo EM for Local Refinement **10**
6.1 The EM objective . . . 10
6.2 E-step . . . 11
6.3 M-step . . . 11
6.4 Convergence . . . 11
6.5 MCEM as a refinement stage . . . 11

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7 Model Selection **11**
7.1 AIC . . . 11
7.2 Bootstrap variance of $\widehat{f}$ . . . 12
7.3 Three-model automated selection . . . 12

8 Stochastic Simulation of Phylogenies **12**
8.1 Forward simulation . . . 12
8.2 Conditional simulation (augmentation) . . . 13
8.3 Batch simulation and survival probability . . . 13

9 Diagnostics **13**
9.1 CEM convergence diagnostics . . . 13
9.2 GAM-based survival probability . . . 13

10 Implementation **14**
10.1 Software architecture . . . 14
10.2 Separation of concerns . . . 14
10.3 Parameter layout . . . 15

11 Simulation Studies **15**
11.1 Estimation power . . . 15
11.2 Model selection power . . . 15

12 Summary of Control Parameters **16**

13 Mathematical Appendix **17**
13.1 Derivation of the log-sum-exp identity . . . 17
13.2 ESS in log-space . . . 17
13.3 Annealing decay rate . . . 17

Abstract

emphasis is an R package for studying species diversification and its ecological drivers. It implements a general birth–death model in which per-lineage speciation and extinction rates depend on arbitrary state covariates — current species richness, total phylogenetic diversity, and lineage-specific evolutionary time. Because the marginal likelihood over the unobserved extinct lineages is analytically intractable, the package uses *importance sampling on augmented trees*: a proposal distribution augments the observed extant tree with stochastically drawn extinct lineages, and the resulting Monte Carlo estimate of the marginal log-likelihood serves as the objective for two complementary optimisers — a Monte Carlo Cross-Entropy Method (CEM) for global search and a Monte Carlo EM (MCEM) algorithm for local refinement. Model selection uses the Akaike Information Criterion computed from the IS log-likelihood estimate at the optimum. This document gives a self-contained mathematical derivation of every component.

1 Notation

Table 1 summarises the principal symbols used throughout.

Table 1: Principal notation.

Symbol	Meaning
$\mathcal{T}$	Observed extant tree (phylo object / branching-time vector)
$\mathcal{T}^+ = (\mathcal{T}, z)$	Augmented tree: $\mathcal{T}$ together with stochastically drawn extinct lineages z
$\mathbf{t} = (t_1 > t_2 > \cdots > t_n > 0)$	Ordered branching times of the extant tree
$\boldsymbol{\theta} \in \mathbb{R}^p$	Parameter vector of the diversification model
$\lambda(s, t; \boldsymbol{\theta})$	Instantaneous speciation rate of lineage s at time t
$\mu(s, t; \boldsymbol{\theta})$	Instantaneous extinction rate of lineage s at time t
$N(t)$	Number of extant lineages at time t
$P(t)$	Total pendant branch-length (sum of time since last split) at t
$E(s, t)$	Evolutionary pendant time of lineage s: time since its last speciation
$p(\mathcal{T}, \mathcal{T}^+ \mid \boldsymbol{\theta})$	Complete-data likelihood
$q(z \mid \mathcal{T}, \boldsymbol{\theta})$	Proposal density for the extinct lineages
$\ell_i^{(f)}$	$\log p(\mathcal{T}, z_i \mid \boldsymbol{\theta})$ — model log-joint
$\ell_i^{(q)}$	$\log q(z_i \mid \mathcal{T}, \boldsymbol{\theta})$ — proposal log-density
w_i	IS log-weight: $w_i = \ell_i^{(f)} - \ell_i^{(q)}$
S	IS sample size (augmented trees per particle)
M	CEM population size (particles per iteration)
$\widehat{f}(\boldsymbol{\theta})$	IS estimate of $\log p(\mathcal{T} \mid \boldsymbol{\theta})$
ESS	Effective sample size

2 The Diversification Model

2.1 General rate specification

The emphasis model belongs to the family of state-dependent birth–death processes [Etienne et al., 2012, Janzen et al., 2015]. Every lineage s alive at time t (measured forward from the crown) is subject to a per-lineage speciation rate and extinction rate that depend on the current macroecological state of the clade:

$$
\begin{aligned}
\lambda(s,t;\boldsymbol{\theta}) &= f(\beta_0 + \beta_N N(t) + \beta_P P(t) + \beta_E E(s,t))\,, \\
\mu(s,t;\boldsymbol{\theta}) &= f(\gamma_0 + \gamma_N N(t) + \gamma_P P(t) + \gamma_E E(s,t))\,,
\end{aligned} \tag{1}
$$

where the full parameter vector is $\boldsymbol{\theta} = (\beta_0, \beta_N, \beta_P, \beta_E, \gamma_0, \gamma_N, \gamma_P, \gamma_E)^\top \in \mathbb{R}^8$.

State covariates. The three covariates in (1) capture distinct ecological signals:

$N(t)$ *Species richness.* The number of lineages currently alive. Negative β_N produces diversity-dependent suppression of speciation (ecological saturation), while positive β_N encodes positive diversity feedbacks.

$P(t)$ *Phylogenetic diversity (PD).* The sum of pendant branch-lengths at time t:

$$
P(t) = \sum_{s:\text{alive at } t} E(s,t). \tag{2}
$$

PD measures the ecological breadth available to the clade; high PD with low N may permit faster diversification.

$E(s,t)$ *Evolutionary pendant time.* The time elapsed since the lineage s last diverged. Lineages with large E have had longer to accumulate reproductive isolation, making them more likely to speciate (positive β_E) or, if the environment has changed, to go extinct (positive γ_E).

Link function. The link function $f : \mathbb{R} \to [0, \infty)$ ensures non-negative rates:

$$
f(\eta) = \begin{cases} \max(0,\eta) & \text{linear link (default)}, \\ \exp(\eta) & \text{exponential link.} \end{cases} \tag{3}
$$

The linear link is piecewise-linear and allows rates to reach exactly zero; the exponential link guarantees strictly positive rates at the cost of less interpretable coefficients.

2.2 Special cases (model shortcuts)

By setting subsets of $\{\beta_N, \beta_P, \beta_E, \gamma_N, \gamma_P, \gamma_E\}$ to zero, equation (1) reduces to well-known models (Table 2).

The number of free parameters is always $p = 2 + 2k$ where $k \in \{0, 1, 2, 3\}$ is the number of active covariates. The exponential link is not supported for the EP covariate because the resulting integrals lack a closed form (see Section 3).

Table 2: Model shortcuts and their free parameters.

String	Formula	Active covariates	Free parameters
`"cr"`	~ 1	none	(β_0, γ_0)
`"dd"`	$\sim N$	$N(t)$	$(\beta_0, \beta_N, \gamma_0, \gamma_N)$
`"pd"`	$\sim \mathrm{PD}$	$P(t)$	$(\beta_0, \beta_P, \gamma_0, \gamma_P)$
`"ep"`	$\sim \mathrm{EP}$	$E(s,t)$	$(\beta_0, \beta_E, \gamma_0, \gamma_E)$
—	$\sim N + \mathrm{PD}$	$N(t), P(t)$	$(\beta_0, \beta_N, \beta_P, \gamma_0, \gamma_N, \gamma_P)$
—	$\sim N + \mathrm{PD} + \mathrm{EP}$	$N(t), P(t), E(s,t)$	all 8 parameters

3 The Phylogenetic Likelihood

3.1 Observed versus augmented data

The observed data $\mathcal{T}$ consists of the extant tree: branching times $\mathbf{t}$ and the number of surviving lineages n. Because the birth–death process generates extinct lineages that are not observed, the *marginal* likelihood

$$p(\mathcal{T} \mid \boldsymbol{\theta}) = \int p(\mathcal{T}, z \mid \boldsymbol{\theta}) \, \mathrm{d}\nu(z) \tag{4}$$

integrates over the space of all possible extinct-lineage histories z, with ν a dominating measure on augmented-tree space. This integral is analytically tractable only for the constant-rate (CR) and some diversity-dependent (DD) models; for EP and general covariate models it is intractable.

3.2 Complete-data log-likelihood

Given an augmented tree $\mathcal{T}^+ = (\mathcal{T}, z)$ with K total lineages (extant and extinct), lineage k alive on the interval $[a_k, d_k]$ where a_k is its birth time and d_k its death time (or the crown age for extant lineages), the complete-data log-likelihood of the birth–death process is

$$\log p(\mathcal{T}, z \mid \boldsymbol{\theta}) = \underbrace{\sum_{k=1}^{K} \sum_{\text{events of } k} \log r_k(t_{\text{event}})}_{\text{event contribution}} - \underbrace{\sum_{k=1}^{K} \int_{a_k}^{d_k} \Big[\lambda(k,t;\boldsymbol{\theta}) + \mu(k,t;\boldsymbol{\theta})\Big] \, \mathrm{d}t}_{\text{survival contribution}}, \tag{5}$$

where $r_k(t)$ is the rate of whatever event (speciation or extinction) terminates lineage k at time t. Specifically:

1. Each speciation event of lineage k at time t contributes $\log \lambda(k,t;\boldsymbol{\theta})$.
2. Each extinction event of lineage k at time t contributes $\log \mu(k,t;\boldsymbol{\theta})$.

3. The integrated hazard for lineage k over its lifetime is

$$H_k(\boldsymbol{\theta}) = \int_{a_k}^{d_k} \Big[\lambda(k, t; \boldsymbol{\theta}) + \mu(k, t; \boldsymbol{\theta})\Big] \, \mathrm{d}t. \tag{6}$$

Closed-form integrals. For the linear link, the integral $H_k(\boldsymbol{\theta})$ has a closed form for all supported covariate combinations. For the EP covariate with exponential link, the integral $\int \exp(\gamma_0 + \gamma_E E(k, t)) \, \mathrm{d}t$ requires $E(k, t) = t - a_k$, giving $\int_{a_k}^{d_k} e^{\gamma_0 + \gamma_E (t - a_k)} \, \mathrm{d}t$ which is analytically tractable, but combining EP with other exponential-link covariates is not currently supported.

3.3 The proposal distribution

The proposal $q(z \mid \mathcal{T}, \boldsymbol{\theta})$ draws extinct lineages conditioned on the observed extant tree $\mathcal{T}$ using a *thinning* (rejection sampling) approach on a Poisson process with rate bounded by $\lambda_{\max}$:

1. For each inter-event interval of the extant tree, extinct-lineage candidates are generated as events of a homogeneous Poisson process with rate $\lambda_{\max}$.
2. Each candidate time t^* is accepted as an actual extinction event with probability $\lambda(t^*; \boldsymbol{\theta})/\lambda_{\max}$.
3. Accepted extinctions initiate new lineages that are recursively simulated until they go extinct before the present.

The resulting log-proposal probability is

$$\log q(z \mid \mathcal{T}, \boldsymbol{\theta}) = \log q_{\text{aug}}(z \mid \mathcal{T}) + \text{correction for thinning acceptance probabilities}, \tag{7}$$

and is computed analytically during the simulation so that it is available alongside $\ell^{(f)}$ for importance-sampling reweighting.

4 Importance-Sampling Likelihood Estimation

4.1 The IS estimator

Let $z_1, \ldots, z_S \overset{\text{iid}}{\sim} q(z \mid \mathcal{T}, \boldsymbol{\theta})$. The marginal likelihood (4) can be written as the expectation

$$p(\mathcal{T} \mid \boldsymbol{\theta}) = \mathbb{E}_q \left[\frac{p(\mathcal{T}, z \mid \boldsymbol{\theta})}{q(z \mid \mathcal{T}, \boldsymbol{\theta})} \right], \tag{8}$$

which is estimated without bias by

$$\widehat{L}_S(\boldsymbol{\theta}) = \frac{1}{S}\sum_{i=1}^{S} \exp\Big(\ell_i^{(f)} - \ell_i^{(q)}\Big) = \frac{1}{S}\sum_{i=1}^{S} \exp(w_i), \tag{9}$$

where $w_i = \ell_i^{(f)} - \ell_i^{(q)}$ are the IS log-weights. The IS log-likelihood estimate is

$$\widehat{f}(\boldsymbol{\theta}) = \log \widehat{L}_S(\boldsymbol{\theta}) = \log\left(\frac{1}{S}\sum_{i=1}^{S} \exp(w_i)\right). \tag{10}$$

Equation (10) is evaluated in numerically stable form using the log-sum-exp identity:

$$\widehat{f}(\boldsymbol{\theta}) = w_{\max} + \log\left(\frac{1}{S}\sum_{i=1}^{S} \exp(w_i - w_{\max})\right), \quad w_{\max} = \max_i w_i. \tag{11}$$

Remark 1 (Jensen's inequality bias)**.** *Because* log *is concave, Jensen's inequality gives* $\widehat{f}(\boldsymbol{\theta}) = \log \mathbb{E}_q[\exp(w)] \le \mathbb{E}_q[w]$*. The estimator* $\widehat{f}$ *is therefore a downward-biased estimate of* $\log p(\mathcal{T} \mid \boldsymbol{\theta})$*, though the bias vanishes as* $S \to \infty$*.*

4.2 Bias-corrected estimator

The exact identity relating $\log \mathbb{E}[X]$ to the cumulants of $\log X$ yields a moment-based correction. Let $\ell_{\text{hat}} = \frac{1}{S}\sum_i w_i$ and define the central moments

$$m_k = \frac{1}{S}\sum_{i=1}^{S}\Big(w_i - \ell_{\text{hat}}\Big)^k. \tag{12}$$

Then the bias-corrected estimator truncated at order K is

$$\widehat{f}_{\text{BC}}(\boldsymbol{\theta}) = \ell_{\text{hat}} + \log\left(1 + \sum_{k=2}^{K} \frac{m_k}{k!}\right). \tag{13}$$

At $K = 2$ this reduces to the log-normal correction $\ell_{\text{hat}} + \log(1 + m_2/2) \approx \ell_{\text{hat}} + m_2/2$ for small m_2. As $K \to \infty$, $\widehat{f}_{\text{BC}} \to \widehat{f}$ identically; the finite-K truncation is the approximation.

Handling rejected trees. A tree is rejected if augmentation exceeds the limit $\lambda_{\max}$ or the maximum extinct-lineage count. Rejected trees have likelihood $L_i = 0$ (i.e. $w_i = -\infty$) and are excluded from the moment sums but enter the denominator:

$$\widehat{f}(\boldsymbol{\theta}) = \widehat{f}_{\text{valid}}(\boldsymbol{\theta}) + \log\left(\frac{S_{\text{valid}}}{S}\right), \tag{14}$$

where $\widehat{f}_{\text{valid}}$ is computed over the S_{valid} non-rejected trees only.

4.3 Effective Sample Size

The IS effective sample size quantifies how many of the S samples contribute meaningfully to the estimate:

$$\mathrm{ESS} = \frac{\left(\sum_{i=1}^{S} \exp(w_i)\right)^2}{\sum_{i=1}^{S} \exp(2w_i)}. \tag{15}$$

$\mathrm{ESS} \approx S$ indicates roughly uniform weights (high IS efficiency); $\mathrm{ESS} \approx 1$ indicates one tree dominates (IS collapse). We report $\mathrm{ESS}/S \in (0, 1]$ as a diagnostic fraction. For reliable estimation, $\mathrm{ESS}/S \geq 0.1$ is recommended.

4.4 Cross-importance sampling (shared-tree mode)

In shared-tree mode (Mode 2), a single pool of S augmented trees $\{z_1, \ldots, z_S\}$ is drawn at a reference parameter $\boldsymbol{\theta}_0$, and $\widehat{f}$ is evaluated at any $\boldsymbol{\theta}_j$ via

$$\widehat{f}(\boldsymbol{\theta}_j) = \log\left(\frac{1}{S}\sum_{i=1}^{S} \exp\left(\ell^{(f)}(z_i, \boldsymbol{\theta}_j) - \ell_i^{(q)}\right)\right), \tag{16}$$

where $\ell_i^{(q)} = \log q(z_i \mid \mathcal{T}, \boldsymbol{\theta}_0)$ are fixed at simulation time and $\ell^{(f)}(z_i, \boldsymbol{\theta}_j)$ is re-evaluated at the new $\boldsymbol{\theta}_j$ via `eval_logf()`. This avoids re-simulating trees for each particle in the population, at the cost of increased variance when $\boldsymbol{\theta}_j$ and $\boldsymbol{\theta}_0$ are far apart.

5 The Cross-Entropy Method for Parameter Estimation

5.1 Overview

The Cross-Entropy Method (CEM) [Rubinstein, 1999, de Boer et al., 2005] solves the optimisation problem

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \in [\boldsymbol{\ell}, \mathbf{u}]}{\arg\max}\, \widehat{f}(\boldsymbol{\theta}) \tag{17}$$

by maintaining a population of M candidate parameter vectors ("particles") and iteratively concentrating them near the optimum. The key idea is to minimise the KL divergence between the empirical distribution of high-quality particles and the parametric family used to generate them.

5.2 Algorithm

Let $[\boldsymbol{\ell}, \mathbf{u}] \subset \mathbb{R}^8$ be the search box, M the population size, $\rho \in (0, 1)$ the elite fraction, and $\boldsymbol{\sigma}^{(0)} \in \mathbb{R}^8_{>0}$ the initial perturbation standard deviations.

Algorithm 1: Monte Carlo CEM (`emphasis_cem`)

Input: branching times $\mathbf{t}$; bounds $[\boldsymbol{\ell}, \mathbf{u}]$; population size M; elite fraction ρ; IS sample size S; initial SD $\boldsymbol{\sigma}^{(0)}$; annealing step α; max iterations T; plateau tolerance δ; patience P

Output: best parameter estimate $\hat{\boldsymbol{\theta}}$; convergence history

Initialise $\{\boldsymbol{\theta}_j^{(0)}\}_{j=1}^{M}$ uniformly in $[\boldsymbol{\ell}, \mathbf{u}]$
$k \leftarrow 0$, *plateau_count* $\leftarrow 0$
while $k < T$ **do**
  `/* E-step: evaluate IS log-likelihood for every particle */`
  for $j = 1, \ldots, M$ **do**
    Draw $z_1^{(j)}, \ldots, z_S^{(j)} \sim q(\cdot \mid \mathcal{T}, \boldsymbol{\theta}_j^{(k)})$
    Compute $\widehat{f}(\boldsymbol{\theta}_j^{(k)})$ via (10)
  Mark particle j as invalid if $\widehat{f}(\boldsymbol{\theta}_j^{(k)}) = \mathrm{NA}$
  `/* Record best particle */`
  $\hat{\boldsymbol{\theta}}^{(k)} \leftarrow \arg\max_j \widehat{f}(\boldsymbol{\theta}_j^{(k)})$
  $f^{(k)} \leftarrow \widehat{f}(\hat{\boldsymbol{\theta}}^{(k)})$
  `/* Stopping rules */`
  if $\boldsymbol{\sigma}^{(k)} \leq \mathbf{0}$ **then stop** (*annealing*)
  if $k > 0$ *and* $f^{(k)} - f^{(k-1)} < \delta$ **then**
    plateau_count $\leftarrow$ *plateau_count* $+1$
    if *plateau_count* $\geq P$ **then stop** (*plateau*)
  else *plateau_count* $\leftarrow 0$
  `/* M-step: resample around elite particles */`
  Sort particles by $\widehat{f}$; retain top $\lfloor \rho M \rfloor$ elite particles
  Resample $M - \lfloor \rho M \rfloor$ new particles by perturbing elite copies:

$$\boldsymbol{\theta}_j^{(k+1)} = \mathrm{clip}\Big(\boldsymbol{\theta}_{\pi(j)}^{(k)} + \boldsymbol{\epsilon}_j,\ \boldsymbol{\ell},\ \mathbf{u}\Big), \quad \boldsymbol{\epsilon}_j \sim \mathcal{N}(\mathbf{0}, \mathrm{diag}(\boldsymbol{\sigma}^{(k)})^2)$$

  where $\pi(j)$ indexes a uniformly drawn elite particle
  Anneal: $\boldsymbol{\sigma}^{(k+1)} \leftarrow \boldsymbol{\sigma}^{(k)} - \alpha$
  $k \leftarrow k + 1$
$\hat{\boldsymbol{\theta}} \leftarrow \arg\max_k \hat{\boldsymbol{\theta}}^{(k)}$ `// global best across all iterations`

5.3 Stopping rules

Three stopping criteria are checked in order at the end of each iteration:

1. **Annealing exhausted.** $\boldsymbol{\sigma}^{(k)} \leq \mathbf{0}$ element-wise. The population has fully collapsed

onto its current best point; further perturbation would be zero.

2. **Plateau.** The improvement in the best IS log-likelihood, $f^{(k)} - f^{(k-1)}$, has been less than δ (default 10^{-4}) for P (default 5) consecutive iterations. This guards against wasting computation when the surface is flat near the optimum.

3. **Hard cap.** $k = T$ (default $T = 20$). Unconditional maximum-iteration limit.

5.4 Particle rescue under limit exceedance

If all M particles return invalid $\widehat{f}$ (every augmented tree rejected), the limits are escalated geometrically: $\lambda_{\max} \leftarrow 10\lambda_{\max}$, $N_{\max} \leftarrow 10N_{\max}$, and evaluation is retried. If $N_{\max}$ exceeds 10,000 with all particles still invalid, the run is terminated with `stop_reason = "all_failed"` and a warning advising the user to increase `num_points` or widen the bounds.

5.5 Annealing schedule

The initial perturbation SD is set to

$$\sigma_p^{(0)} = \frac{u_p - \ell_p}{4}, \quad p = 1, \ldots, 8, \tag{18}$$

where u_p and ℓ_p are the upper and lower bounds for parameter p. At each iteration the SD is reduced by α:

$$\boldsymbol{\sigma}^{(k+1)} = \boldsymbol{\sigma}^{(k)} - \alpha \mathbf{1}, \quad \alpha = \frac{\sigma_p^{(0)}}{T}. \tag{19}$$

This linear schedule ensures the population collapses smoothly from broad exploration to local exploitation over T iterations.

6 Monte Carlo EM for Local Refinement

6.1 The EM objective

The Monte Carlo Expectation–Maximisation (MCEM) algorithm [Wei and Tanner, 1990, Levine and Casella, 2001] iteratively maximises the expected complete-data log-likelihood

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) = \mathbb{E}_{z \mid \mathcal{T}, \boldsymbol{\theta}^{(t)}} [\log p(\mathcal{T}, z \mid \boldsymbol{\theta})] \,. \tag{20}$$

6.2 E-step

In the E-step, S augmented trees are drawn from $q(z \mid \mathcal{T}, \boldsymbol{\theta}^{(t)})$ and used to form the IS-weighted approximation:

$$\hat{Q}(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) = \frac{\sum_{i=1}^{S} \exp(w_i^{(t)}) \cdot \ell^{(f)}(z_i, \boldsymbol{\theta})}{\sum_{i=1}^{S} \exp(w_i^{(t)})}, \tag{21}$$

where $w_i^{(t)} = \ell^{(f)}(z_i, \boldsymbol{\theta}^{(t)}) - \ell_i^{(q)(t)}$ are the IS log-weights at the current iterate $\boldsymbol{\theta}^{(t)}$, and the denominator normalises them to proper weights.

6.3 M-step

The M-step maximises $\hat{Q}(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)})$ over $\boldsymbol{\theta} \in [\boldsymbol{\ell}, \mathbf{u}]$ using the COBYLA (Constrained Optimisation BY Linear Approximations) algorithm [Powell, 1994] via the `nloptr` interface. Because COBYLA does not require gradient information, it is robust to the Monte Carlo noise in $\hat{Q}$.

6.4 Convergence

MCEM convergence is assessed by monitoring the change in the IS log-likelihood $\widehat{f}(\boldsymbol{\theta}^{(t)})$ across iterations. The algorithm terminates when this change falls below a tolerance ε (default 10^{-1}) for a specified number of consecutive post-burnin iterations. A burn-in phase (default 20 iterations) is excluded from convergence monitoring to allow initial transients to decay.

6.5 MCEM as a refinement stage

Because MCEM converges to a local optimum of $\boldsymbol{\theta} \mapsto \log p(\mathcal{T} \mid \boldsymbol{\theta})$, it is recommended as a *second stage* after the CEM has located a good region of the parameter space:

1. Run CEM to obtain $\hat{\boldsymbol{\theta}}_{\text{CEM}}$.
2. Initialise MCEM at $\hat{\boldsymbol{\theta}}_{\text{CEM}}$ and refine.

7 Model Selection

7.1 AIC

For each fitted model $\mathcal{M}_k$ with p_k free parameters and IS log-likelihood $\widehat{f}(\hat{\boldsymbol{\theta}}_k)$, the Akaike Information Criterion is

$$\text{AIC}_k = -2\,\widehat{f}(\hat{\boldsymbol{\theta}}_k) + 2p_k. \tag{22}$$

The preferred model minimises AIC. The ΔAIC values $\mathrm{AIC}_k - \min_j \mathrm{AIC}_j \geq 0$ quantify the evidence against model k.

Remark 2 (IS bias and AIC)**.** *Because $\widehat{f}$ is a downward-biased estimate of $\log p(\mathcal{T} \mid \hat{\boldsymbol{\theta}})$ (Remark 1), the AIC is upward-biased relative to its true value. When comparing models evaluated at the* same *sample of augmented trees this bias is constant across models and cancels in ΔAIC; when trees are simulated independently per model (the default), residual Monte Carlo noise may affect comparisons with very small ΔAIC.*

7.2 Bootstrap variance of $\widehat{f}$

To quantify IS Monte Carlo uncertainty at the optimum, the package provides a leave-half-out (subsampling) bootstrap variance of $\widehat{f}$:

$$\widehat{\mathrm{Var}}[\widehat{f}(\hat{\boldsymbol{\theta}})] = \frac{1}{B}\sum_{b=1}^{B}\Big(\widehat{f}^{(b)}(\hat{\boldsymbol{\theta}}) - \bar{\widehat{f}}(\hat{\boldsymbol{\theta}})\Big)^2, \tag{23}$$

where $\widehat{f}^{(b)}$ uses a randomly drawn half of the S_{final} augmented trees. A large $\widehat{\mathrm{Var}}$ relative to $|\Delta\mathrm{AIC}|/2$ indicates that the model comparison is noise-dominated and more trees should be used.

7.3 Three-model automated selection

The function `select_diversification_model()` fits the three biologically most common models (DD, PD, EP) in a single call, applies (22), and returns the ranked comparison table together with the best-fit parameter estimates.

8 Stochastic Simulation of Phylogenies

8.1 Forward simulation

Forward simulation of the state-dependent birth–death process proceeds by Gillespie's direct method [Gillespie, 1977]. At each step:

1. Compute the total event rate $R(t) = \sum_s[\lambda(s,t;\boldsymbol{\theta}) + \mu(s,t;\boldsymbol{\theta})]$.
2. Draw the waiting time $\Delta t \sim \mathrm{Exp}(R(t))$.
3. Choose the event type proportional to individual rates.
4. Update the state $(N, P, \{E_s\})$ and the lineage set.

Because the rates λ and μ depend on $N(t)$, $P(t)$, and individual $E(s,t)$ which change continuously, the Gillespie step requires re-computing $R(t)$ at each event. For the linear link with N- and P-dependence the waiting-time distribution still has a closed form; for EP dependence the waiting time is drawn via thinning.

8.2 Conditional simulation (augmentation)

Given an observed extant tree $\mathcal{T}$, conditional simulation draws the extinct lineages z from $q(z \mid \mathcal{T}, \boldsymbol{\theta})$ (Section 3.3). This is the E-step of MCEM and the particle evaluation step of CEM. The C++ engine returns both the augmented tree and $\ell^{(q)}$ so that IS reweighting can be applied immediately.

8.3 Batch simulation and survival probability

Passing a matrix of parameter rows to `simulate_tree()` triggers batch mode: each row is simulated independently, and the fraction of simulations that produce a non-extinct clade estimates the survival probability Pr(clade survives to present $\mid \boldsymbol{\theta}$). This probability is also used to correct the marginal likelihood for conditioning on non-extinction.

9 Diagnostics

9.1 CEM convergence diagnostics

The function `diagnose_cem()` extracts and plots four diagnostic panels from a fitted CEM object:

Log-likelihood trace. $f^{(k)}$ versus iteration k, annotated with the stopping rule. A monotone increasing curve ending at a plateau indicates good behaviour.

Parameter traces. $\hat{\boldsymbol{\theta}}_p^{(k)}$ versus k for each parameter dimension p. Stable late-iteration values indicate that the population has converged.

Population spread. The median and interquartile range of $\widehat{f}(\boldsymbol{\theta}_j^{(k)})$ over all valid particles. Narrowing IQR reflects progressive population collapse.

IS weight distribution. Histogram of w_i at the globally best particle, annotated with ESS. A concentrated histogram near the mean indicates healthy IS weights.

9.2 GAM-based survival probability

For DD models, `train_GAM()` fits a binomial generalised additive model (GAM) to the outcomes of batch forward simulations, estimating Pr(survival $\mid \boldsymbol{\theta}$) as a smooth function of

the parameter space. `predict_survival()` then evaluates this surface at any $\boldsymbol{\theta}$ without re-simulating, enabling fast computation of likelihood corrections for conditioning on non-extinction.

10 Implementation

10.1 Software architecture

emphasis is written in R and C++, with the computationally intensive steps delegated to C++ via Rcpp and parallelised using RcppParallel. The public API consists of five primary functions:

`simulate_tree()` Forward and conditional simulation of birth–death trees; dispatches to the C++ engine.

`augment_trees()` Raw E-step: draw S augmented trees from $q(z \mid \mathcal{T}, \boldsymbol{\theta})$, returning `trees`, `logf`, and `logg`.

`eval_logf()` Re-score pre-computed augmented trees at any $\boldsymbol{\theta}$, enabling cross-IS (Section 4.4).

`estimate_rates()` Unified inference entry point: resolves the model, dispatches to CEM or MCEM, and returns a named `emphasis_fit` object.

`compare_models()` / `select_diversification_model()`
AIC-based model comparison.

10.2 Separation of concerns

The key architectural principle is the separation of simulation and scoring:

1. `augment_trees()` simulates trees and computes both $\ell^{(f)}$ and $\ell^{(q)}$ during the simulation pass.
2. `eval_logf()` re-evaluates $\ell^{(f)}(z, \boldsymbol{\theta}')$ for any new $\boldsymbol{\theta}'$ on already-simulated trees, without re-running the stochastic augmentation.
3. IS aggregation is performed in R via `.is_fhat(logf, logg)`.

This design enables shared-tree mode (Mode 2): a single simulation pass at $\boldsymbol{\theta}_0$ produces trees that are reused for all particles in the CEM population, reducing the total number of augmentations from $M \times S$ to S per iteration, at the cost of higher IS variance.

10.3 Parameter layout

Internally, all functions operate on the full 8-element parameter vector $\boldsymbol{\theta} \in \mathbb{R}^8$ in the order $(\beta_0, \beta_N, \beta_P, \beta_E, \gamma_0, \gamma_N, \gamma_P, \gamma_E)$. Model shortcuts set the inactive elements to zero. The model binary flag $\mathbf{m} = (m_N, m_P, m_E) \in \{0, 1\}^3$ indicates which covariates are active and gates the corresponding rate functions in the C++ likelihood kernel. For the N and P covariates the model flag only affects the proposal (which covariate terms to simulate); for the EP covariate, $m_E = 1$ also switches the rate functions from standard to EP-aware in `eval_logf()`.

11 Simulation Studies

11.1 Estimation power

To evaluate the frequentist accuracy of CEM estimation under the PD model, we simulate $n = 100$ trees from the true parameter vector

$$\boldsymbol{\theta}^* = (\beta_0, \beta_P, \gamma_0, \gamma_P) = (0.6,\ 0.005,\ 0.15,\ -0.003)$$

with crown age $T = 8$, and estimate $\boldsymbol{\theta}$ on each replicate.

For each replicate $r = 1, \ldots, n$ we record the estimate $\hat{\boldsymbol{\theta}}_r$ and the number of extant tips n_r. Point accuracy is summarised by:

$$\text{Bias}_p = \frac{1}{n} \sum_{r=1}^{n} (\hat{\boldsymbol{\theta}}_{r,p} - \boldsymbol{\theta}_p^*), \tag{24}$$

$$\text{RMSE}_p = \sqrt{\frac{1}{n} \sum_{r=1}^{n} (\hat{\boldsymbol{\theta}}_{r,p} - \boldsymbol{\theta}_p^*)^2}. \tag{25}$$

The tree-size effect is visualised by plotting $|\hat{\boldsymbol{\theta}}_{r,p} - \boldsymbol{\theta}_p^*|$ against n_r with a LOWESS smooth. Consistent improvement with n_r confirms that the IS estimator is consistent: larger trees provide a sharper likelihood surface and more informative augmentation.

11.2 Model selection power

We assess the ability of AIC-based selection to recover the true data-generating model in a fully crossed design. Twenty-five trees are simulated from each of the four pure models (CR, DD, PD, EP), giving 100 trees in total. All four models are then fitted to every tree, and the model with the lowest AIC is selected.

The results are summarised in a 4×4 confusion matrix

$$C_{ij} = \#\Big\{\text{trees generated by model } i \text{ for which model } j \text{ was selected}\Big\}, \tag{26}$$

and the row-normalised version $C_{ij}/\sum_j C_{ij}$ (recall matrix). Overall accuracy is $\text{acc} = \text{tr}(C)/\sum_{i,j} C_{ij}$. Per-model recall $C_{ii}/\sum_j C_{ij}$ measures how reliably each true model is identified.

12 Summary of Control Parameters

Table 3: CEM control parameters (`estimate_rates_control("cem")`).

Parameter	Default	Description
`max_iter`	20	Hard iteration cap T
`num_points`	50	Population size M
`sample_size`	1	IS sample size S per particle per iteration
`shared_trees`	`FALSE`	Pool trees across particles (Mode 2)
`disc_prop`	0.5	Elite fraction ρ
`sd_vec`	`NULL`	Initial $\boldsymbol{\sigma}^{(0)}$; auto $= (u - \ell)/4$
`maxN`	10	Max augmentation attempts before rejection
`max_missing`	10^4	Max extinct lineages per augmentation
`max_lambda`	500	Maximum $\lambda_{\max}$ for thinning
`tol`	10^{-4}	Plateau tolerance δ
`patience`	5	Consecutive plateau iterations P
`bias_correct`	`FALSE`	Apply moment-based bias correction
`n_boot`	0	Bootstrap replicates for $\widehat{\text{Var}}[\widehat{f}]$
`num_threads`	1	Parallel threads for particle evaluation

Table 4: MCEM control parameters (`estimate_rates_control("mcem")`).

Parameter	Default	Description
`sample_size`	200	IS sample size S per EM iteration
`tol`	0.1	Convergence tolerance ε
`burnin`	20	Burn-in iterations excluded from convergence check
`max_missing`	10^4	Max extinct lineages per augmentation
`max_lambda`	500	Maximum $\lambda_{\max}$ for thinning
`num_threads`	1	Parallel threads

13 Mathematical Appendix

13.1 Derivation of the log-sum-exp identity

For numerical stability, (10) is never evaluated as $\log(\frac{1}{S}\sum_i \exp(w_i))$ directly because the exponentials can overflow or underflow. Instead, factor out $w_{\max} = \max_i w_i$:

$$\begin{aligned}\log\left(\frac{1}{S}\sum_{i=1}^{S} e^{w_i}\right) &= \log\left(\frac{e^{w_{\max}}}{S}\sum_{i=1}^{S} e^{w_i - w_{\max}}\right) \\ &= w_{\max} + \log\left(\frac{1}{S}\sum_{i=1}^{S} e^{w_i - w_{\max}}\right). \end{aligned} \tag{27}$$

Since $w_i - w_{\max} \leq 0$, all terms $\exp(w_i - w_{\max}) \in (0, 1]$ are safe to evaluate.

13.2 ESS in log-space

From the normalised weights $\tilde{w}_i = \exp(w_i)/\sum_j \exp(w_j)$, the ESS (15) equals $1/\sum_i \tilde{w}_i^2$. In log-space:

$$\mathrm{ESS} = \frac{\left(\sum_i e^{w_i}\right)^2}{\sum_i e^{2w_i}} = \exp[2\,\mathrm{lse}(\mathbf{w}) - \mathrm{lse}(2\mathbf{w})]\,, \tag{28}$$

where $\mathrm{lse}(\mathbf{v}) = \max_i v_i + \log\sum_i e^{v_i - \max_j v_j}$.

13.3 Annealing decay rate

If $\boldsymbol{\sigma}^{(0)} = c\mathbf{1}$ for some $c > 0$ and $\boldsymbol{\sigma}^{(k+1)} = \boldsymbol{\sigma}^{(k)} - \alpha\mathbf{1}$, the population collapses (all $\sigma_p^{(k)} \leq 0$) at iteration $k^* = \lceil c/\alpha \rceil$. Setting $k^* = T$ gives the default annealing step $\alpha = c/T = \sigma_p^{(0)}/T$.

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